

AgentLab

A research lab run by agents

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github.com/shuds13/AgentLab

What it is

You give an agent a goal and a system. It runs the investigation itself.

A lab run by agents

A campaign agent submits work to an HPC system, reads what comes back, decides what to try next, and keeps going until it has an answer — for hours or days, with nobody in the loop.

One campaign per question

Each investigation is a campaign with its own goal, task definition and records. Campaigns are independent and run side by side, sharing the framework and the system definitions.

Visible and steerable

Any number of campaigns, all watchable from one Slack channel, and redirectable while they run. Nobody stands up new infrastructure per question.

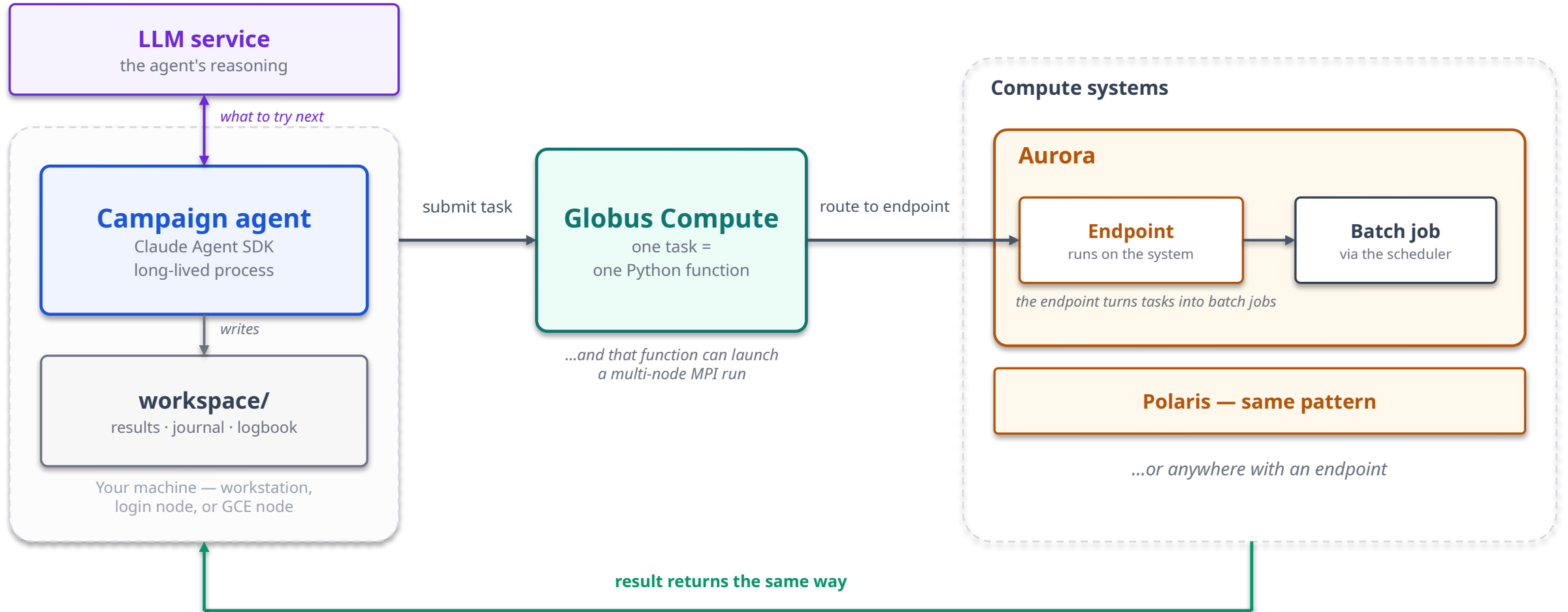
github.com/shuds13/AgentLab

built on the same machinery as github.com/shuds13/cas-framework, which coordinates many agents on a single search campaign

Runs anywhere the Claude Agent SDK and a Globus Compute endpoint can be reached — the agent does not need to be on the HPC system.

How a campaign runs

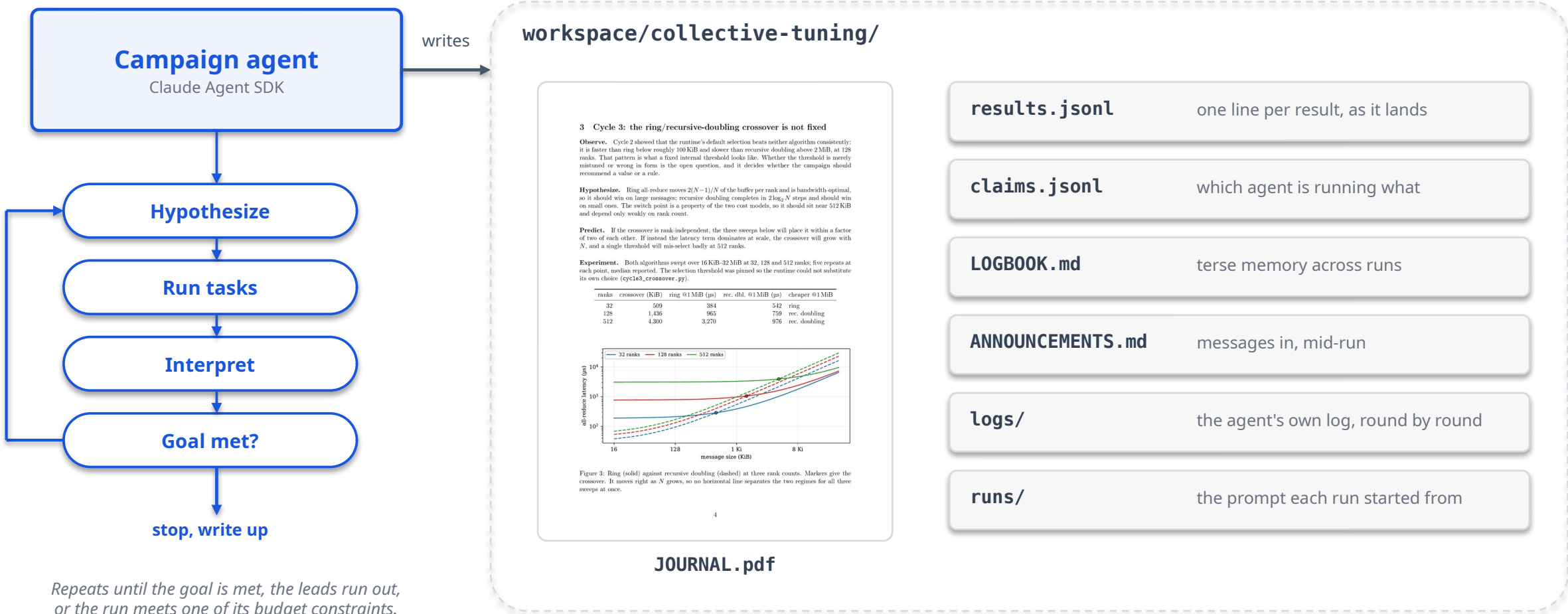
The agent runs user-side and submits work to remote systems.



The loop is the point: submit, read, decide, submit again — for hours or days, without a person in it.

The workspace

One directory per campaign. Everything the agent produces lands in it.

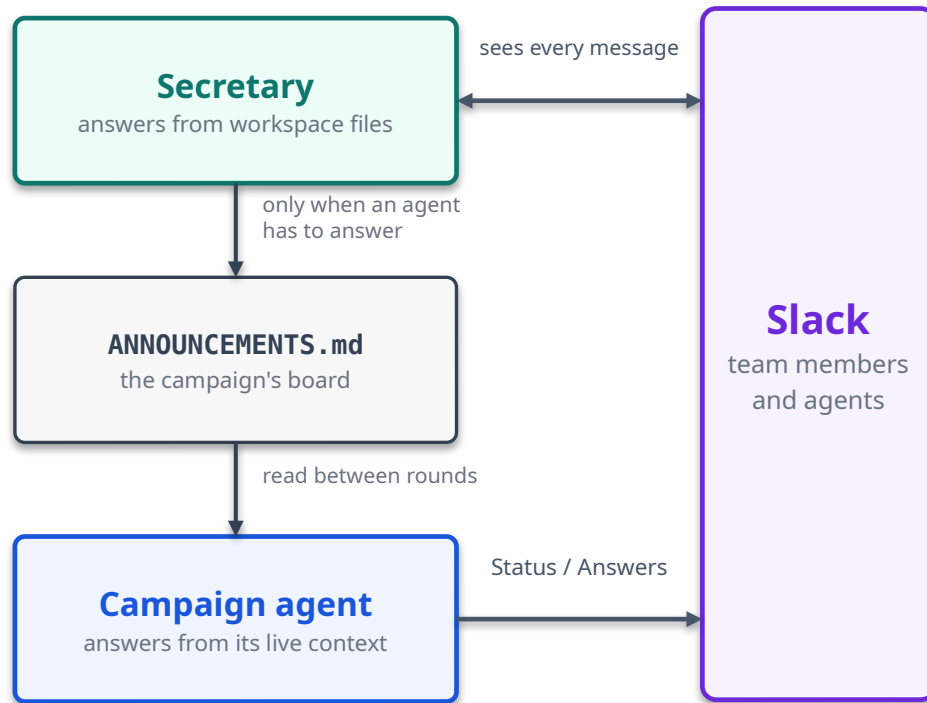


Repeats until the goal is met, the leads run out, or the run meets one of its budget constraints.

Several agents can share one workspace — `claims.jsonl` is how they avoid running the same work twice.

Human in the loop

The lab runs on its own. You can still watch it, question it, and redirect it — from Slack.



You can put an announcement in ANNOUNCEMENTS.md for the agents at any time.

The screenshot shows a Slack channel with three messages. The first message from **AgentLab** at 12:58 PM reports that the **aurora1** agent is offline and provides a command to start it. The second message from **AgentLab** at 1:04 PM reports that **aurora1** has started. The third message is a question from user **shudson** at 1:11 PM asking about the prompt used by the **Aurora** agent. A red 'New' indicator is next to the last message.

AgentLab APP 12:58 PM
[v-opt/aurora1] 🚨 Agent exiting -- Globus Compute endpoint 'agentlab-aurora' is not online (status=offline). Start it: globus-compute-endpoint start agentlab-aurora --detach

AgentLab APP 1:04 PM
[v-opt/aurora1] 🚀 Agent aurora1 started — v-opt on aurora · claude-opus-5[1m] · critic gpt-sol.

shudson 1:11 PM
What prompt is the Aurora agent using New

AgentLab APP 1:11 PM
[secretary] aurora_20260828_130455 (started today 13:04, host compute-386-03, claude-opus-5[1m]) is running `user_prompt_hpc.md` with the campaign's standard `prompt.md` — the same brief as yesterday's run.

an alert, a start notice, a question, and the secretary answering it

The secretary answers from the workspace files; the research agents answer from their live context.

Built on the Claude Agent SDK

Everything that makes Claude Code powerful is available in the SDK.

What comes with it

- Built-in tools
- Subagents
- Hooks
- MCP
- Permissions
- Sessions
- Skills, commands and memory
- Plugins

LiteLLM

Easily deploy a proxy for multi-vendor models.

A session is written to disk, so a run can be picked up later

```
bin/list_agents.sh --all
```

 lists the session ids

Fork a session from interactive Claude Code

Work the problem by hand, then hand the session to a campaign agent.

```
RESUME_SESSION=<session-id> ./run.sh
```

Run a postmortem with the research agent

Reopen a finished campaign's session and ask what happened.

```
claude -r <session-id>
```

Getting started

Clone the repository, start your agent in it, and ask.

1 Clone it

```
git clone  
https://github.com/shuds13/AgentLab
```

2 Start your agent in it

```
cd AgentLab && claude
```

3 Say

"Help me set up."

What you need

- Claude Agent SDK, and an LLM service it can reach
- Python 3.10+, and the claude CLI on PATH and authenticated
- Access to a compute system, and a project to charge

Where to run

- Preferably a shared persistent node, so the team reaches the same campaigns
- Inside tmux. A campaign runs for hours or days, and closing the terminal kills it with jobs still in flight

AGENTS.md in the repository root is the onboarding script — the agent reads it and walks you through the rest.